

module2day2

1. Layer 7 – Application

Main job: Provides network services to applications.

Examples: HTTP, HTTPS, DNS, FTP, SSH.

Example:

You open Firefox and visit `google.com`.



Layer 7 = Application/network service



2. Layer 6 – Presentation

Main job: Handles **data format, encryption, and compression**.

Example:

When you visit a website using HTTPS, your data is encrypted before being sent.



Layer 6 = Format + Encrypt + Compress

3. Layer 5 – Session

Main job: Establishes and manages communication sessions.

Example:

You log into an online service and maintain a communication session while using it.



Layer 5 = Manage the session



4. Layer 4 – Transport ☆

Main job: Provides end-to-end communication.

Main protocols:

- **TCP** → reliable
- **UDP** → faster, no connection setup

Example:

When downloading a file:



TCP makes sure the data arrives reliably.

For a live voice call:



UDP is commonly preferred because speed is important.



Layer 4 = TCP/UDP + Port numbers



5. Layer 3 – Network ☆☆☆

Main job: Uses **IP addresses** and determines how packets travel between networks.

Device: Router

Example:

Your PC:

192.168.1.10



Router



Internet



Google Server



The router uses IP addresses to determine where the packet should go.



Layer 3 = IP + Routing + Router



6. Layer 2 – Data Link ☆☆☆

Main job: Uses **MAC addresses** and creates **frames** for local network communication.

Device: Switch

Example:

PC-A — Switch — PC-B



The switch looks at MAC addresses to decide where to forward the frame.



Layer 2 = MAC + Frame + Switch



7. Layer 1 – Physical

Main job: Transmits raw **bits (0 and 1)** using electrical, light, or radio signals.

Examples:

- Ethernet cable
- Fiber optic
- Wi-Fi radio
- Hub
- Repeater

Example:

101010101010



Electrical signal through cable

Copy



Layer 1 = Bits + Signals + Media



One Real-Life Example: Opening YouTube

You type:

youtube.com

The data goes through the layers:



Layer 7 → HTTPS / DNS
Layer 6 → Encryption
Layer 5 → Session management
Layer 4 → TCP/UDP + Ports
Layer 3 → IP addresses
Layer 2 → MAC addresses + Frames
Layer 1 → Electrical/radio signals

At the sender:



Data → Segment → Packet → Frame → Bits

At the receiver, the process is reversed:



Bits → Frame → Packet → Segment → Data

This is called **encapsulation and decapsulation**.



The 4 Most Important Layers for Cisco

Layer	Remember	Device
L1 Physical	Bits	Hub
L2 Data Link	MAC / Frame	Switch
L3 Network	IP / Packet	Router
L4 Transport	TCP/UDP / Ports	—

Easy example

If your PC cannot connect to another PC:

- **Cable problem?** → L1
- **Wrong VLAN/MAC problem?** → L2
- **Wrong IP/default gateway?** → L3
- **Port/TCP problem?** → L4

That's why the OSI model is very useful for **network troubleshooting**.